What is machine learning?
a) A type of programming language
b) A software framework for building applications
c) A subset of artificial intelligence that enables machines to learn from data
d) A tool for data visualization
Correct answer: c

What are the three primary components of a machine learning model?
a) Data, algorithm, and hardware
b) Algorithm, data, and evaluation metrics
c) Model, training data, and testing data
d) Hardware, software, and data
Correct answer: b

What is the term for the process of feeding data into a machine learning model to train it?
a) Inference
b) Training
c) Validation
d) Testing
Correct answer: b

What type of machine learning problem involves predicting a continuous output value based on input features?
a) Supervised classification
b) Unsupervised clustering
c) Regression
d) Time series forecasting
Correct answer: c

What is the term for the process of selecting the most relevant features to use in a supervised machine learning model?
a) Feature engineering
b) Dimensionality reduction
c) Regularization
d) Hyperparameter tuning
Correct answer: b

Which algorithm is commonly used for supervised classification problems, especially when dealing with imbalanced datasets?
a) Logistic Regression
b) Decision Trees
c) Support Vector Machines (SVMs)
d) Random Forests
Correct answer: c

What type of machine learning problem involves identifying patterns or structures in unlabeled data?
a) Supervised clustering
b) Unsupervised classification
c) Clustering
d) Dimensionality reduction
Correct answer: c

Which algorithm is commonly used for unsupervised clustering problems, especially when dealing with high-dimensional datasets?

a) K-Means
b) Hierarchical Clustering
c) DBSCAN
d) Principal Component Analysis (PCA)
Correct answer: a

What is the term for the process of reducing the number of features in a dataset to improve model interpretability and prevent overfitting?
a) Feature selection
b) Dimensionality reduction
c) Regularization
d) Hyperparameter tuning
Correct answer: b

Which algorithm is commonly used for unsupervised dimensionality reduction, especially when dealing with high-dimensional datasets?
a) PCA (Principal Component Analysis)
b) t-SNE (t-Distributed Stochastic Neighbor Embedding)
c) Autoencoders
d) Generative Adversarial Networks (GANs)
Correct answer: a

What is the term for the slope of the regression line in linear regression?
a) Coefficient
b) Intercept
c) Residual
d) R-Squared
Correct answer: a

Which algorithm is commonly used for linear regression problems, especially when dealing with simple datasets?
a) Ordinary Least Squares (OLS)
b) Ridge Regression
c) Lasso Regression
d) Elastic Net Regression
Correct answer: a

What is the term for the process of reducing the impact of outliers in linear regression by using robust loss functions?
a) Regularization
b) Outlier detection
c) Robust regression
d) Hyperparameter tuning
Correct answer: c

Which algorithm is commonly used for linear regression problems, especially when dealing with multiple features and interactions between them?
a) Generalized Linear Model (GLM)
b) Bayesian Linear Regression
c) Non-Parametric Regression
d) Neural Networks
Correct answer: a

What type of machine learning problem involves predicting a binary output value based on input features?
a) Supervised classification

b) Unsupervised clustering
c) Regression
d) Time series forecasting
Correct answer: a

Which algorithm is commonly used for logistic regression problems, especially when dealing with simple datasets?
a) Logistic Regression (LR)
b) Decision Trees
c) Support Vector Machines (SVMs)
d) Random Forests
Correct answer: a

What is the term for the process of reducing the impact of overfitting in logistic regression by using regularization techniques?
a) Regularization
b) Outlier detection
c) Robust regression
d) Hyperparameter tuning
Correct answer: a

Which algorithm is commonly used for logistic regression problems, especially when dealing with multiple features and interactions between them?
a) Generalized Linear Model (GLM)
b) Bayesian Logistic Regression
c) Non-Parametric Logistic Regression
d) Neural Networks
Correct answer: b

What is the primary difference between regression and classification machine learning problems?
a) The type of output value predicted
b) The type of input features used
c) The type of data used for training
d) The algorithm used for prediction
Correct answer: a

Which problem involves predicting a continuous output value, while classification involves predicting a discrete output value?
a) Regression
b) Classification
c) Clustering
d) Dimensionality reduction
Correct answer: a

What is the term for the process of evaluating the performance of a machine learning model on unseen data?
a) Model validation
b) Model testing
c) Model deployment
d) Model monitoring
Correct answer: b

Which algorithm is commonly used for hyperparameter tuning, especially when dealing with

complex models and large datasets?
a) Grid Search
b) Random Search
c) Cross-Validation
d) Bayesian Optimization
Correct answer: d

What is the term for the process of selecting the most relevant features to use in a machine learning model?
a) Feature selection
b) Dimensionality reduction
c) Regularization
d) Hyperparameter tuning
Correct answer: a

What type of neural network architecture involves multiple layers, each with its own set of learnable weights and biases?
a) Feedforward Neural Network (FNN)
b) Convolutional Neural Network (CNN)
c) Recurrent Neural Network (RNN)
d) Autoencoder
Correct answer: b

Which algorithm is commonly used for deep learning problems, especially when dealing with images and computer vision tasks?
a) AlexNet
b) VGGNet
c) ResNet
d) Inception Network
Correct answer: c

What is the term for the process of optimizing the weights and biases in a deep neural network using backpropagation?
a) Forward Propagation
b) Backward Propagation
c) Stochastic Gradient Descent (SGD)
d) Adam Optimization
Correct answer: b

What type of neural network architecture involves multiple layers, each with its own set of learnable weights and biases?
a) Feedforward Neural Network (FNN)
b) Convolutional Neural Network (CNN)
c) Recurrent Neural Network (RNN)
d) Autoencoder
Correct answer: a

Which algorithm is commonly used for neural network problems, especially when dealing with complex datasets and multiple features?
a) Multilayer Perceptron (MLP)
b) Long Short-Term Memory (LSTM)
c) Gated Recurrent Unit (GRU)
d) Generative Adversarial Network (GAN)
Correct answer: a

What is the term for the process of selecting the most relevant features to use in a neural network?
a) Feature selection
b) Dimensionality reduction
c) Regularization
d) Hyperparameter tuning
Correct answer: a

Which algorithm is commonly used for neural network problems, especially when dealing with large datasets and complex models?
a) Stochastic Gradient Descent (SGD)
b) Adam Optimization
c) RMSProp
d) Adagrad
Correct answer: b